arcOS
are you **QRV?**

Field Manual

22.1.0-2 “Acadia” Release
Revised: 2025-06-03

Welcome to arcOS!

The Amateur Radio Community Operating System

INTRODUCTION:

The Amateur Radio Community Operating System is a bootable live Linux distribution installed on a USB drive, and it is built to "just work" with the affordable Digirig device. arcOS is founded on the belief that digital communications within communities of operators can be accessible and easy to use for ALL, regardless of license class or experience.

HARDWARE REQUIREMENTS/RECOMMENDATIONS:

USB Drive (min. 16GB, USB 3.0)
Laptop/Desktop Computer (min. 8GB RAM)
Digirig Mobile + Cable* (Lite and DR891 NOT supported)
Transceiver
GPS Receiver (optional, but STRONGLY recommended)

*Note: While some popular transceivers are equipped with an internal soundcard. **arcOS depends on the use of a Digirig Mobile** device as the computer-to-radio interface. Many of the transceivers with internal soundcards are also supported by one or more Digirig cables. The following list is not exhaustive. Please conduct your own research to find an appropriate cable for your transceiver.*

IC-7300 – Uses the Icom IC-706 cable
FT-991A – Uses the Kenwood TS-480 cable
IC-705 – NO KNOWN SOLUTION FOR USE WITH DIGIRIG

INCLUDED AMATEUR RADIO SOFTWARE:

Direwolf 1.7 (Packet Modem)
ARDOP 1.0.4.1.3 (Soundcard Modem)
Pat 0.16.0 (Winlink Client)
YAAC 1.0-beta209 (APRS Client)
FL-Suite (Multi-mode Soundcard Modem and Additional Tools)
    FLDigi 4.1.20, FLMsg 4.0.23, FLAmp 2.2.09, FLRig 2.0.04, FLArq 4.3.8
WSJT-X 2.7.0 (FT8 Client)
JS8Call 2.2.0 (JS8 Client)
Paracon 1.1.0 (Packet Client)
QSSTV 9.5.8 (Slow Scan Television Client)
*VARA HF 4.8.9 (Soundcard Modems for Windows)
*VARA FM 4.3.8 (Soundcard Modems for Windows)
Hamlib 4.5.5 (Rig Control)

**Requires purchase of an unlock key*

CAT/RIG CONTROL:

While arcOS does include FLRig and `rigctl[d]`, and users may save customized settings to take advantage of CAT control, **NO DEVELOPER SUPPORT FOR CAT/RIG CONTROL WILL BE PROVIDED**. This is for the following reasons:

1) CAT control isolates the operator from their own equipment, often exposing a lack of proficiency when it is not available.

2) CAT control introduces extra complexity with regard to troubleshooting, with which many inexperienced users aren't prepared to engage.

3) Attempting to maintain support for every past, present, and future transceiver is not reliably feasible.

arcOS strives to help operators become proficient with digital modes, without creating "appliance" operators.

SUPPORT:

IRC: arcOS includes Hexchat ("Menu > Internet > Hexchat) as an IRC client for chat support (or just to hang out and chat with other users). Hexchat is configured to automatically connect to the #arcOS-Linux channel on the Libera.Chat network, and uses the operator callsign as the "nickname" for the user. Come say hi!

Note: Developer presence in the chat channel is not guaranteed.

GitHub: If you encounter something in arcOS that you believe to be a bug, please consider starting a "Discussion" topic on GitHub.

https://github.com/kg4vdk/arcos-linux-modules/discussions

If the issue is determined to be a bug, an "Issue" ticket can be opened. When reporting issues, please be as detailed as possible. Providing remote support for users is EXTREMELY difficult if a reported issue does not contain information useful for replicating/troubleshooting the issue.

Information to include in any report of a suspected bug:

FULL SYSTEM REPORT – Generate a link to your system report using "Menu > System Info", and select "Upload system information". A browser window will open with the text of your system report. DO NOT copy/paste the report. Instead, copy the link from the browser address bar, and include the link in your report.

Include the QRV Modules version, along with information about any user customizations that have been made.

Information about your station equipment in use
Provide the radio and Digirig version you are using.

Describe the bug
A clear and concise description of what the bug is.

Steps to reproduce the behavior:

1. Go to '...'
2. Click on '....'
3. Scroll down to '....'
4. See error

Expected behavior
A clear and concise description of what you expected to happen.

What actually happened
Include any errors, verbatim.

Screenshots
If applicable, add screenshots to help explain your problem.

Additional context
Add any other context about the problem.

Please DO NOT use Winlink for support requests!

PERSISTENT STORAGE:

The first time arcOS is booted after USB creation, any free space on the USB device is configured as an exFAT filesystem to be used as persistent storage. Any files not saved in this partition will be lost when the system is powered off/rebooted. For a single user, the persistent storage is organized as follows:

```
ARCOS-DATA
   |-Downloads (Common to all users)
   |-QRV
      |-LOGS (Common to all users)
      |-OFFLINE-MAPS (Common to all users)
      |-CALLSIGN (Specific to the configured user)
         |-SAVED (Base configurations and QRV Profile configurations)
         |-arcos-linux-modules (Updates provided via GitHub)
         |-LOGS (Specific to the configured user)
            |-CORE, COMMUNITY, and USER Modules
```

Suggested items to keep in persistent storage:

Checklists for station setup
Radio manuals (your own, plus any common ones that could benefit others)
Frequently used forms
Communication plans (ICS-205)
Frequency lists (including frequently used Winlink gateways)
Contact information

STATION SETUP:

When arcOS is booted, users will be presented with the “Station Setup” window. After supplying some basic information about the operator/station, arcOS will be configured for “out-of-the-box” use with any transceiver compatible with the Digirig device. For users outside the United States, please treat the “State” field appropriately for the locale (e.g. province, territory, etc.). Once the station is ready for use, the user will receive a notification that “CALLSIGN is QRV!”

Note: "QRV" is Amateur Radio shorthand for "I am ready!"

The most recent information passed to "Station Setup" is saved for use on each reboot or run of "Station Setup", so users will only need to change fields if necessary.

Auto-Configuration:

Once configured to your liking, you can set arcOS to automatically configure your station with the information saved by "Station Setup". To enable auto-configuration, simply copy your station configuration from `/ARCOS-DATA/.operators/station-info_$CALLSIGN` to a hidden file named `.autoconfig` in `/ARCOS-DATA`.

QRV MODULES:

QRV Modules are included in the ISO image, and are updated occasionally with fixes, improvements, and features. When an update for the QRV Modules is available, users will be notified by the appearance of an icon (⟳) in the "System Information" display at the bottom right of the desktop window. To update the QRV Modules, use "Menu > arcOS Tools > Update QRV Modules." When the "Station Setup" information is provided, the scripts provided by the QRV Modules are run in the following order:

1) CORE Modules
2) COMMUNITY Modules
3) USER Modules

CORE MODULES:

The CORE Modules provide basic functionality for the included Amateur Radio software. They should not be modified by users, and any user-made changes will be overwritten when the modules are updated.

COMMUNITY MODULES:

The COMMUNITY Modules are built and maintained by community members, and can provide useful functionality beyond what is included in arcOS. Changes to these modules will also be overwritten when the QRV Modules are updated. No COMMUNITY Modules are "active" by default. Each module in this category contains a module script in its directory (`00_MODULE-NAME.sh`). To activate a COMMUNITY Module, simply move the module script UP one level. For example:

```
COMMUNITY
   |-EXAMPLE
      |-00_EXAMPLE.sh (NOT ACTIVE, inside EXAMPLE directory)

COMMUNITY
   |-00_EXAMPLE.sh (ACTIVE, up one level from EXAMPLE directory)
   |-EXAMPLE
```

Activated COMMUNITY Modules are "remembered" during a QRV Module update, and re-activated once the modules are up to date. Users can re-number the module scripts, in the case that particular ordering is desired.

USER MODULES:

USER Modules can be used to personalize the system appearance, customize system preferences, install software, or do just about anything a user wants to do at the "Station Setup" runtime. This can be especially useful if users pre-download software packages, and install them with a script in the USER Modules directory.

Note:
One of the final CORE modules restarts the Cinnamon desktop environment (this is why the screen flashes back momentarily during Station Setup). If your USER (or COMMUNITY) module requires running before the Cinnamon restart, simply make sure the module script includes "_PRE_" in its filename.

Example: 00_PRE_MY-USER-MODULE.sh

PERSONALIZATION:

The desktop background image or slideshow can be personalized by creating a directory named "`ARCOS-DATA/QRV/CALLSIGN/Wallpapers`". For a static background image, place a JPEG image in this directory, named "`wallpaper.jpg`." For a slideshow of images, create a sub-directory named "`Slideshow`", and place the desired images within it.

The applications pinned to the panel are remembered per user.

QRV PROFILES:

The last field of the "Station Setup" window allows for selection of a "QRV Profile." This is useful for deploying saved configurations for differing scenarios. Most of the configurations for the included Amateur Radio software are savable via right-clicking the application icon in the bottom panel (or via "Menu > arcOS Tools > SAVE APPNAME CONFIG"). When saving an application's configuration, users will be presented with the option to input a custom "QRV Profile" name. If no custom name is provided, the configuration will be saved to a profile named "DEFAULT". Once a profile has been created, it will be available in the dropdown menu during "Station Setup." To switch between profiles, it is usually sufficient to simply re-run the "Station Setup" from the main menu.

Example:

A user saves their Pat configuration (with password) to the "DEFAULT" profile. They also save their YAAC configuration (with icon, beacon message, and SSID) to a profile named "MOBILE". When they run "Station Setup" and select "MOBILE", YAAC will be setup using this "MOBILE" profile, Pat will use the "DEFAULT" profile configuration, and all other application will use the base configuration as shipped in arcOS.

PERSISTENT SOFTWARE:

For packages not included in arcOS, but available as a deb file, users can utilize the "`download-packages`" command in a terminal to cache the packages in persistent storage for installation at boot time, before Station Setup runs. Prior to using this tool, users should test installing the packages manually using "`apt`". Once any dependency issues are resolved, pass the package list to the command as follows:

```
$ download-packages package_1 [package_2 package_3]
```

Each time the command is used, the downloaded packages are stored in /ARCOS-DATA/QRV/.packages/packages-$MD5HASH. This allows for users to remove some packages without disturbing other groups of packages.

BACKUP & RESTORE:

To backup the currently configured operator, use the “Menu > arcOS Tools > Backup Operator” utility. This will create a backup of the current operator’s files. The utility will prompt for a location to save the backup. It is recommended to save the backup onto a storage device other than the arcOS persistent storage. The “OFFLINE-MAPS” are NOT included in the backup.

To restore an operator from a backup file, ensure any `.autoconfig` file has been deleted, and reboot the system. At the “Select Operator” screen, select “Restore from backup”. A drag-and-drop window will be presented. Open the ARCOS-DATA drive on the Desktop to access the file browser. Locate the backup file, and drag it into the window. When the restoration is complete, the Station Setup will be presented, configured for the restored operator.

The backup and restore functions are designed to be used as a recovery tool, not as a migration tool between releases. When restoring a backup from an older release into a newer release, you may need to remove the arcos-linux-modules directory from the backup before restoring. This can be safely done by making a copy of your target backup, opening it in “Archive Manager”, browsing to your callsign directory, and deleting the arcos-linux-modules directory. This will usually allow for your saved configurations to be migrated, and any user modules you’d like to recover will remain in your original backup for retrieval.

UNLOCK KEYS:

In order for VARA modems to function on Linux systems, Wine is required to emulate a Windows environment. In addition to being designed for Windows, VARA is NOT open source. Considerable effort is involved in ensuring reliable functioning of this software in arcOS. For users desiring to use the VARA modems, an unlock key is available for purchase at https://arcos-linux.com. **Without an unlock key, Wine and VARA modems WILL NOT function**. All other software will function without an unlock key. If you have purchased an unlock key, you will receive a confirmation email notifying you your callsign has been added. Once you have received this email, use “Menu > arcOS Tools > Download arcOS Keys” to download the key file to your persistent storage.

Note: If the “Station Setup” is run before the key file is present on the system, a reboot is required to allow arcOS to validate the supplied callsign at “Station Setup” runtime. Only downloading the key file requires an internet connection. Once the key file is downloaded, validation takes place on the local system. Installation of VARA modems, likewise, do not require an internet connection.

VARA REGISTRATION:

During the initial VARA installation process, users are prompted to enter their VARA Registration Code. If a registration code is not provided, VARA will operate in a “trial mode.” If a user later desires to purchase a registration code for VARA, it can be entered in the “REGISTRATION_CODE” file located in the `/ARCOS-DATA/QRV/CALLSIGN/SAVED/VARA` directory.

NETWORKS:

Configurations for wi-fi networks can be saved using "Menu > arcOS Tools > SAVE WIFI CONNECTION". If a saved wi-fi connection is available, arcOS will automatically connect to it. This can also be used to save information like static IP addresses.

GPS:

arcOS should work with any GPS device which is recognized by `gpsd` (3.22). If your device is recognized (indicated by "Waiting for GPS lock…" in the Station Information on the desktop, but you have difficulty obtaining a position fix, please consider the following possibilities and reading through these informative links:

- An unobstructed view of the sky is paramount.
- If "cold-starting", the GPS almanac can take a minimum of 12.5 minutes to be downloaded (often longer).
- "Cold" starts are any time the GPS almanac has become stale. This is often due to significant changes in location since last use, or being disused for a long period of time.
- https://gpsd.gitlab.io/gpsd/faq.html#startup
- https://en.wikipedia.org/wiki/Time_to_first_fix

By default, arcOS is configured to use an attached GPS as a time source if no network time server is available. If GPS is actively being used as the time source, a clock icon will be displayed next to the coordinates and gridsquare in the System Information display at the bottom right of the desktop.

SOUND:

arcOS can save the preferred system and Digirig sound levels on a per machine basis (using the serial number of the machine on which arcOS is booted). To adjust the system sound levels, simply use the speaker icon in the system tray (bottom right panel area). To adjust the Digirig sound levels, you can open a terminal and use alsamixer. Press F6 to select the "USB PnP Sound Device", then use the up/down arrow keys to adjust the levels for the "Speaker" and the "Mic" (labeled "Capture"). Once you have set the audio levels as desired, press "ESC" to exit `alsamixer`. To save the sound settings persistently, use "Menu > arcOS Tools > SAVE SOUND CONFIG".

PAT WINLINK:

Save Password: To easily save the Winlink password for your callsign, right-click on the "PAT WLNK" icon in the bottom panel and select "SET PAT WLNK PASSWORD". This will set the password for this session. To save the password persistently, right-click the "PAT WLNK" icon and select "SAVE PAT WLNK CONFIG", and choose the "DEFAULT" profile or another profile name as appropriate for your use case. If you choose not to save your Winlink password, you'll simply be prompted to provide it when connecting to a gateway.

Connect to a Gateway (Send/Retrieve Messages): Make a connection to a Winlink gateway by selecting "Action > Connect" from the top menu. Then, choose the desired transport method, and ensure the relevant modem is running. Input the desired target gateway callsign and SSID. Select "Connect".

RMS List: The RMS List is a record of active Winlink gateways, which provides the callsign, distance from your station, mode, and frequency used by the gateway.

Update RMS List: arcOS saves the RMS list in persistent storage, and you can update it by using “Action > Connect”, “Show RMS list”, and selecting “Update cache”.

Position Reports/Requests: To **send a position report** in Pat, select “Action > Position” from the top menu. If a GPS device is attached, and has a location fix, the latitude and longitude will be auto-populated. Network based geolocation can provide coordinates, however the accuracy is dependent on your ISP. Otherwise, you can manually enter your location’s coordinates. Include a comment relevant to your scenario. Once ready, select “Post”, and the position report will be placed in your Pat outbox.

To **request position reports** sent by other stations, you have a few options. Each of these methods requires composing a simple message. Using the top menu in Pat, select “Action > Compose”.

Request 30 Nearest Reports from the Last 10 Days:

```
TO: INQUIRY
SUBJ: REQUEST
BODY: WL2K_NEARBY
```

Request Last Report from a Single Station:

```
TO: QTH
SUBJ: POSITION REQUEST
BODY: CALLSIGN
```

Request Last *N* Reports from a Single Station:

```
TO: QTH
SUBJ: POSITION REQUEST
BODY: CALLSIGN N
```

Once your request message is composed, follow the steps above to connect to a Winlink gateway.

Peer-To-Peer (P2P): In arcOS, Pat is configured by default to “listen” for P2P connections on each of the included modems. P2P connections, as the name suggests, do not require the use of a gateway. In order to send/receive a P2P message, the other station must also be configured properly. Only the callsign of the other station is needed (do not include “@winlink.org”). When composing the message, ensure that the “P2P Only” box is checked. Select the desired transport method (must be in use by both stations), and enter their callsign as the target, and select “Connect”.

YAAC:

Offline maps for use in YAAC can be downloaded from within YAAC. From the main menu in YAAC, select “File > OpenStreetMap > Download Pre-Imported Tiles”. If a GPS is connected, and has a valid location fix, YAAC will automatically use your position as the center point for the specified radius. If no GPS is available, you can manually enter your desired coordinates.

FLDIGI:

By default, FLDIGI is configured to operate using the MT63-2KL mode, which is suitable for HF/VHF/UHF use, including use over repeaters (obtain the permission of the repeater owner before using digital modes).

FLDIGI is also configured to respond to “pings” from other stations. This can be useful for groups of operators wishing to operate a persistent net. To ping another station, use the “Ping” macro button in FLDIGI, which will place “PING>” in the transmit window. Append the callsign of the station to be pinged, and then use the “TX >|” macro button to send the ping and automatically return to receiving. If the pinged station is on frequency with FLDIGI running, it will receive a pop-up notification, and respond with a timestamped “pong” containing the current status of FLAMP, FLMSG, and the system’s uptime.

HAMRS:

Due to the frequency of updates to HAMRS, it is not included in arcOS. However, a CORE Module exists, which will allow for persistent use of HAMRS once a user downloads the AppImage.

To use HAMRS system wide, simply download the AppImage from the developer’s website, and place it in `ARCOS-DATA/QRV/LOGS`. Ensure the AppImage is named `hamrs.AppImage`.

To use HAMRS for a specific user, simply download the AppImage from the developer’s website, and place it in `ARCOS-DATA/QRV/CALLSIGN/LOGS`. Ensure the AppImage is named `hamrs.AppImage`.

When opening HAMRS, you may receive a notification that an update is available. If you wish to use the updated version, you may download the new AppImage from the HAMRS website, and place it in the appropriate directory, and the next time “Station Setup” runs, the new version of HAMRS will be deployed.

FILE SHARING:

Warpinator is included in arcOS for file sharing over a local network. Warpinator is configured with the group code “arcOS” to enable the ability to utilize compression when sharing files. If sharing files with non-arcOS users, you may need to change the group code for the session. You can do this from within Warpinator by using the menu icon (top left), and selecting “Preferences”, and changing the Group Code on the “Connection” tab.

DESKTOP SHARING:

X11VNC provides the ability to share/control your arcOS machine from another arcOS machine, or any machine with a compatible VNC client. Desktop sharing can be enabled/disabled using “Menu > arcOS Tools > Enable/Disable Desktop Sharing”. A notification will be displayed with the address to use on the remote machine for the connection. In addition, an icon will be placed next to the IP address in the Station

Information display on the bottom right of the desktop to indicate that Desktop Sharing is active.

VIKING MAPPING:

Viking is a useful mapping tool for viewing and editing maps, tracks, and waypoints. By default the application is configured to utilize the maximum 4GB map cache size. Any maps downloaded by Viking are cached in the OFFLINE-MAPS directory for persistent use.

PRIVACY:

GnuPG keys are persistently saved per user. Kleopatra is included as a key manager and tool. Veracrypt is also included. Users should be mindful that on multi-user systems, other users COULD gain access to your persistent GnuPG keys. Take appropriate precautions.

KNOWN ISSUES:

FLDIGI:

Occasionally, high CPU usage may occur when running FLDIGI. This is due to a bug in the Cinnamon desktop environment and/or its interaction with the fltk toolkit used by FLDIGI. If this occurs, you can restart the desktop environment by pressing CTL+ALT+ESC. The desktop will flash and then reload. This is a shortcut provided by the Linux Mint development team to address this issue in Cinnamon.

ARDOP:

If the ARDOP modem is closed by clicking the “x” on the terminal window, ARDOP will continue to run in the background. This may cause other modems to fail to start. If this occurs, re-open ARDOP, and press “CTL+C” in the terminal window. This will cleanly kill the ARDOP process.

VARA HF:

If VARA HF is opened, closed, and re-opened in quick succession, you will receive an error. This is due to VARA HF holding a port open until it times out. The timeout is 60 seconds. Simply wait 60 seconds or longer before re-opening VARA HF to prevent this issue.

PAT WINLINK:

If a form is opened to compose a message, but then closed without submitting the form, you may receive a “form instance key not valid” error. To remedy this, simply close the PAT WINLINK window, and re-open it. You may want to copy your composed message text to a file to prevent the need to type it again.

NOTES